GitHub Copilot as a Coding-Agent Harness

Agentic development in VS Code, the Copilot CLI, and GitHub.com

Xebia

Welcome

What This Session Is

- A hands-on tour of GitHub Copilot as a **coding-agent harness**
- One agent loop, one set of tools and context files, one approval model
- Three places it runs: VS Code, the terminal, and GitHub.com
- You'll use all three on a real .NET or Java codebase
- We finish by comparing Copilot with OpenCode and Kiro

Agenda

Time	Block
0:00	Welcome and your practice repo
0:10	Module 1 — The harness and chat in VS Code
1:00	Module 2 — Agent mode, instructions, agents, MCP
1:50	Module 3 — The cloud agent and code review
2:20	Break
2:30	Module 3 activity, part 2
2:50	Module 4 — The Copilot CLI
3:40	Conclusion and questions

How Each Module Runs

- 25 minutes of live demo on the **library** app
- 25 minutes of hands-on work on the **orders** app, in your own repo
- Each activity links its module's slides, for review whenever you like
- You won't finish every task, and that's by design
- Each activity has a `solution/` folder for afterward

Two Apps, Two Languages

App	What it does	Who uses it
Library	Books, loans, due dates, late fees	The instructor's demos
Orders	Catalog, validation, discounts, tax	Your activities

- Each app exists in **.NET 10** and **Java 21**
- Pick the language you work in every day

Your Environment

- A VM with VS Code, Git, the Copilot CLI, the .NET SDK, and a JDK
- The GitHub account you were given, with a Copilot seat
- Check it now (clone the courseware first if it isn't there):

```
git clone https://github.com/coding-agent-harness-class/courseware $HOME\copilot-course
cd $HOME\copilot-course\courseware\setup
powershell -ExecutionPolicy Bypass -File .\setup.ps1 -VerifyOnly
```

- Then sign in: Copilot in VS Code, `copilot` then `/login`, and `gh auth login`

Make Your Practice Repo (1)

- Enterprise accounts can't fork from outside the enterprise, so you **copy** the app
- Your copy is a **private** repo in your own account
- Its Actions tab stays empty: personal repos of enterprise accounts don't run Actions
- Step by step: `courseware/setup/student-quick-start.md`
- Sign in `gh` first, with the `workflow` scope so you can push workflows:

```
gh auth login --web --git-protocol https --scopes workflow
gh auth setup-git
```

Make Your Practice Repo (2)

```
$me = gh api user -q .login
$repo = "$me/orders-dotnet"
git clone https://github.com/coding-agent-harness-class/orders-dotnet $HOME\orders-dotnet
Set-Location $HOME\orders-dotnet
gh repo create $repo --private
git remote remove origin
git remote add origin "https://github.com/$repo.git"
git push -u origin main
gh repo set-default $repo
```

- Java: `orders-java` everywhere it says `orders-dotnet`
- Post your repo's link in the chat when it opens with `gh repo view --web`

One Thing to Expect

- Copilot won't give you exactly what it gave the person next to you
- Different runs and different models take different paths
- The skill is reading what the agent did, judging it, and steering it
- Activity steps describe what you should **see**, not exact output

Let's Begin

- Courseware: `README.md` at the root of the repository
- Activities: `courseware/activities/`
- Reference reading: `courseware/docs/`
- Questions any time, in the chat or out loud